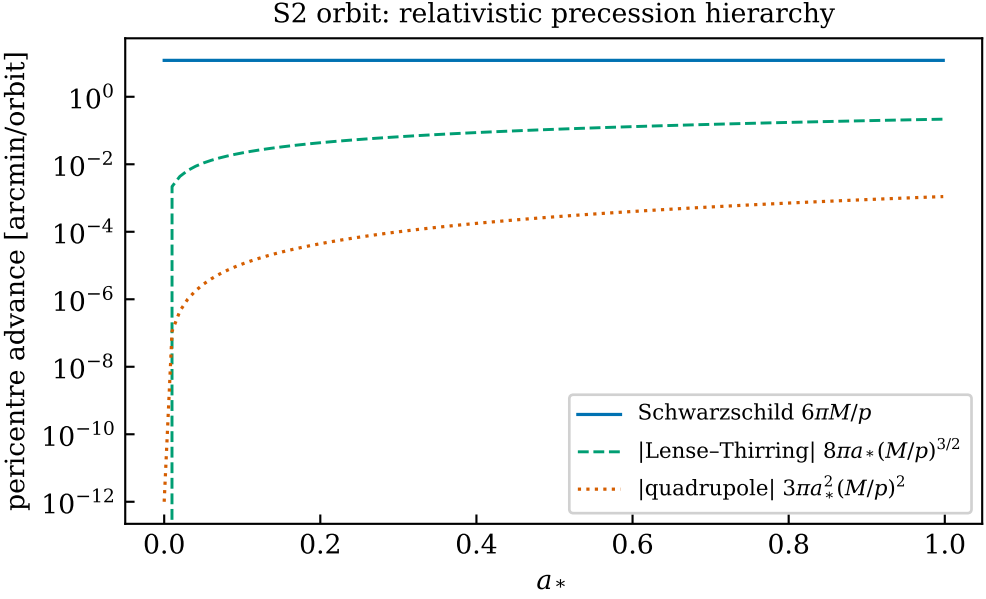
S2 orbit: relativistic precession hierarchy
pericentre advance [arcmin/orbit]
$10^{0}$
$10^{-2}$
$10^{-4}$
$10^{-6}$
$10^{-8}$
$10^{-10}$
$10^{-12}$
0.0
0.2
0.4
0.6
0.8
1.0
$a_*$
Schwarzschild $6\pi M/p$
|Lense–Thirring| $8\pi a_*(M/p)^{3/2}$
|quadrupole| $3\pi a_*^2(M/p)^2$